BRAC UNIVERSITY **B**

Department of Computer Science and Engineering

Examination: Semester Final Duration: 1 Hour 50 Minutes	Semester: Summer 2025 Full Marks:30

CSE 260: Digital Logic Design

Question 4 is mandatory. Answer any 2 questions from Question 1, 2, 3. Show calculations and mention MSB, LSB where required. Figures in the right margin indicate marks. **Understanding the question is part of the exam.**

1.CO3 a) Let's assume you have come across a memory system that uses a Memory Address Register (MAR) with 10 flip-flops and a Memory Buffer Register (MBR) with 48 flip-flops. [3]

I. Determine the data input size and output size of this memory.
II. How many address lines are required to access the entire memory?
III. What is the total memory capacity of the system in KiloBytes (KB)?

b) At BRAC University, during course registration, many students may request the same course simultaneously. To prevent system overload and ensure fairness, the university uses a Course Request Prioritizer (CRP) to decide which student's request will be processed first. [7]

Each student sends a digital request flag for a course:
F3 → Request from a graduating student
F2 → Request from a student with backlog in that course
F1 → Request from a regular student
F0 → Request from a freshman trying to pre-enroll

The request flags work as follows:
1 = Request is active
0 = No request from that category

Priority Rules:

- If no one requests the course (all 0s) → Output: *Don't Care*.
- If the combined request number (F3F2F1F0) is divisible by 3 → It is treated as a department-handled bulk allocation. Priority goes to MSB.
- If exactly three categories of students request at the same time → It's considered a special high-demand case. In this situation, priority is given to the middle groups (first F2, then F1), since they represent students with backlogs or regular enrollment needs.
- In all other situations → It's treated as a general competition for seats. Here, priority goes to the LSB.

Design the circuit diagram of a Priority Encoder that will take the four inputs (F3, F2, F1, F0) representing student requests for a course and provide the output based on the priority rules described above.

2. CO3 a) **Implement** the following function using both 4x1 mux[s] and 2x1 mux[s] in a single circuit: [4]
$F(A,B,C,D,E) = \Sigma(1,3,4,7,9,13,15,18,20,23,25,26,31)$
Make sure that you use the lowest number of mux[s] possible. Make sure that your circuit contains more 4x1 mux[s] than 2x1 mux[s].

b) Suppose, you are building a circuit using encoder and decoder that takes a 3 bit number as input and outputs the excess 3 version of that number. Once the result [X] is found, then you expand the circuit using X and another 4 bit number Y based upon the following conditions: [6]

- If X is divisible by 4 and Y is greater than or equal to 4, then it performs X-Y
- Performs X+Y otherwise

NB: You have to **design** the whole circuit combining the initial and the extended part both.

3.CO3 A delivery robot at BRAC University follows a weekly delivery schedule visiting four locations: **[10]**

- Library (L)
- Cafeteria (C)
- Lab (B)
- Admin Office (A)

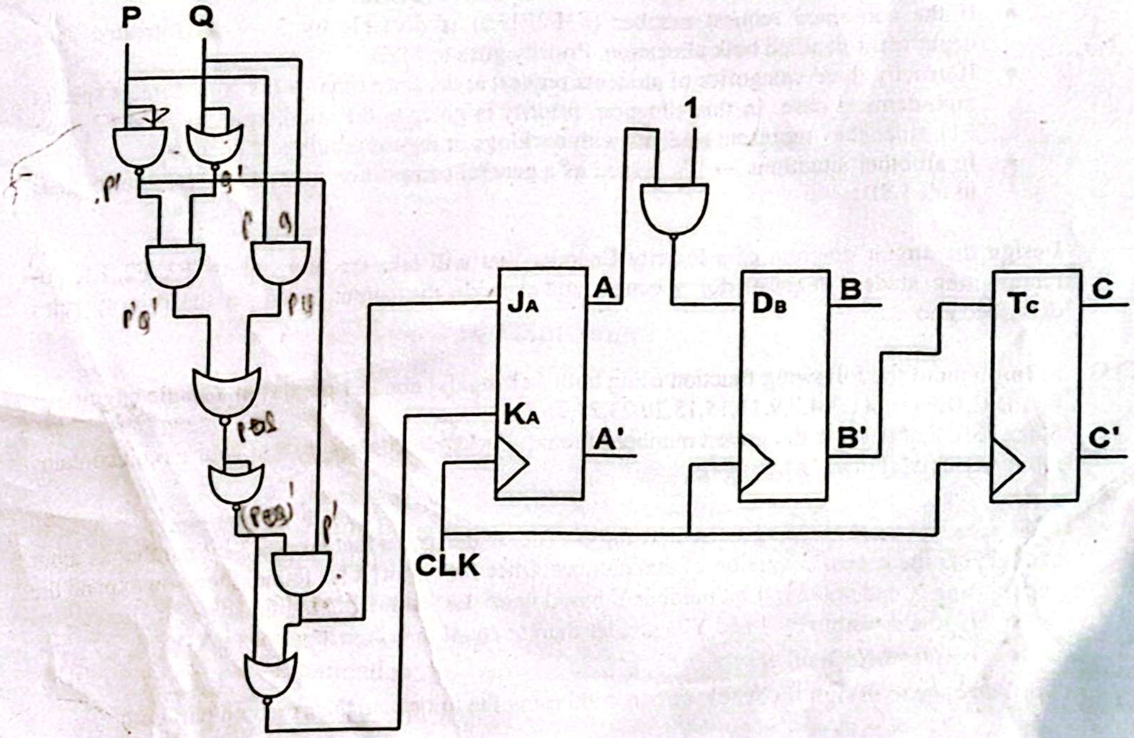
The robot follows this delivery sequence:

1. Starts at the Library (L) to pick up books. L
2. Moves from Library → Cafeteria (C) to drop off meals. L→C
3. Moves from Cafeteria → Lab (B) to deliver lab equipment. C→B
4. Returns from Lab → Cafeteria (C) to pick up extra drinks. B→C
5. Moves from Cafeteria → Library (L) to return for the next batch of books. C→L
6. Moves from Library → Admin Office (A) to submit reports. L→A
7. Moves from Admin Office → Lab (B) for optional equipment pickup. A→B
8. Moves from Lab → Cafeteria (C) for a final check. B→C
9. Moves from Cafeteria → Admin Office (A) to finalize deliveries. C→A
10. Moves from Admin Office → Library (L) for regular service. A→L

Design a counter based on the above scenario using SR flip-flops.
Use Don't Care conditions for any unused states/transitions. Certain transitions may occur more than once. Follow the sequence as described without confusion or assumption.

4. CO3 a) The high time in a clock cycle is 30 ms and the low time is 28 ms. Calculate the frequency [in Hz] and duty cycle for that signal? **[3]**

b) Draw timing diagrams [positive edge triggered] for J_A, K_A, A, D_B, B, B', C. Answer in the separate sheet given with the question. **[7]**



2. **CO3**

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Make sure that you use the lowest number of mux[s] possible. Make sure that your circuit contains more 4x1 mux[s] than 2x1 mux[s].

[6]

b) Suppose, you are building a circuit using encoder and decoder that takes a 3 bit number as input and outputs the excess 3 version of that number. Once the result [X] is

found, then you expand the circuit using X and another 4 bit number Y based upon the following conditions:

- If **X** is divisible by 4 and **Y** is greater than or equal to 4, then it performs **X-Y**
- Performs **X+Y** otherwise

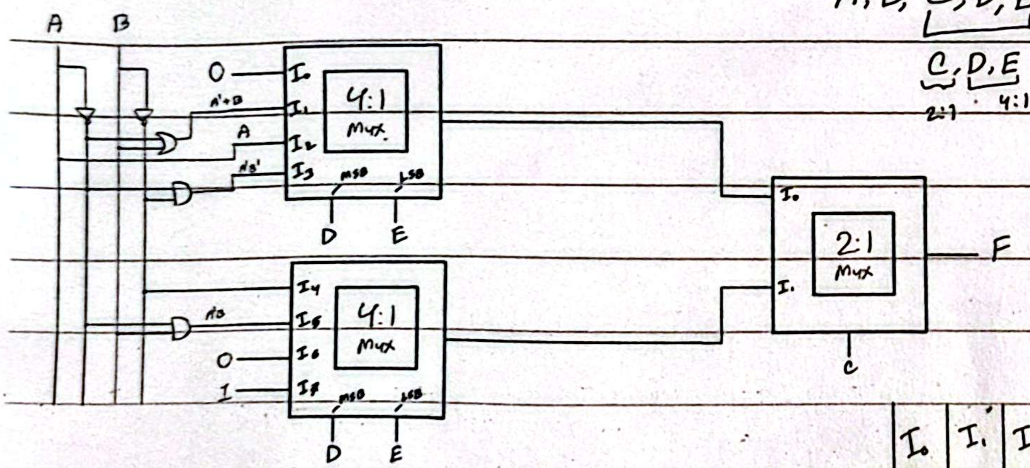
NB: You have to **design** the whole circuit combining the initial and the extended part both.

Solution-

MSB LSB

$A, B, \underbrace{C, D, E}$

$\underbrace{C}_{2:1}, \underbrace{D, E}_{4:1}$



	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$A'B'$	0	1	2	3	4	5	6	7
$A'B$	8	9	10	11	12	13	14	15
AB'	16	17	18	19	20	21	22	23
AB	24	25	26	27	28	29	30	31
	0			$A'B'$	B'	$A'B$	0	1

$A'+B \leftarrow A'B'+A'B+AB$

$= A'+AB$

$AB'+AB \rightarrow A$